Product-Market Fit Measurement Engine

Based on Superhuman's framework (First Round Review) + Sean Ellis's PMF Survey + Sequoia Arc

The Problem

Everyone talks about product-market fit. Almost nobody can measure it. "You'll know it when you feel it" is terrible advice. This framework gives you an exact number, an exact process, and an exact improvement loop.

Step 1: The Sean Ellis Survey

Ask users who have experienced the core of your product (used it at least twice in the past two weeks) this question:

> ***"How would you feel if you could no longer use [product]?"*** *- A) Very disappointed - B) Somewhat disappointed - C) Not disappointed*

The Benchmark: If 40%+ answer "Very disappointed," you have product-market fit. Below 40%, you don't.

After benchmarking nearly 100 startups, Sean Ellis found this to be the single most reliable leading indicator. Companies below 40% almost always struggled. Companies above it almost always found growth.

Step 2: Three More Questions

After the first question, ask:

Q2: “What type of people do you think would most benefit from [product]?” → This reveals your real target audience (often different from who you think).

Q3: “What is the main benefit you receive from [product]?” → This reveals your actual value proposition (often different from what you’re marketing).

Q4: “How can we improve [product] for you?” → This reveals the roadmap — but filter it (see Step 4).

Step 3: Segment and Score

Don’t look at the aggregate. Segment your responses:

1. **Who are your “Very Disappointed” users?** Profile them. This is your ideal customer persona — not who you assumed, but who the data says.
2. **What do the “Very Disappointed” users say the main benefit is?** This is your positioning. Use their words, not yours.
3. **Where do the “Somewhat Disappointed” users overlap with the “Very Disappointed” users?** This is your expansion opportunity. The “Somewhat” users are almost there — what would tip them over?

Step 4: The Improvement Loop

This is Superhuman’s actual process (repeated every 2-4 weeks):

A. Categorize the Q4 feedback

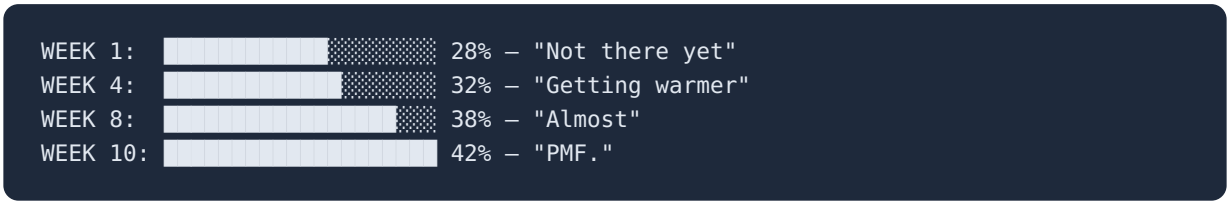
Split Q4 responses into two buckets: - Feedback from “Very Disappointed” users → **Do this.** These people already love you. Make them love you more. - Feedback from “Not Disappointed” users → **Ignore this.** They were never your customer. Building for them dilutes your product.

B. Prioritize the roadmap

For every feature request from “Very Disappointed” users, ask: - Does this deepen the core benefit? → High priority - Does this add a new benefit? → Medium priority - Does this fix a minor annoyance? → Low priority

C. Re-measure every 2-4 weeks

Run the survey again. Track the % over time. Your goal: move from wherever you are toward 40%.



Step 5: Sequoia's Four "Terrifying Questions"

Once you have early signal, stress-test it with Sequoia's Arc framework. Before you scale, answer these honestly:

1. "Do I have Hair-on-Fire PMF, Hard-Fact PMF, or Future-Vision PMF?"

- **Hair-on-Fire:** Users have an urgent problem and your product is the fire extinguisher. They'll tolerate a bad product because the pain is so acute. (Example: early Uber in cities with terrible taxi service)
- **Hard-Fact:** Users switch because of measurably superior economics or performance. They need proof, not promises. (Example: Stripe's 7-line integration vs. competitors' 7-API-call integration)
- **Future-Vision:** Users believe in a new way of doing things and adopt early despite friction. Hardest to achieve, highest ceiling. (Example: early Tesla buyers)

2. "Who is my desperate customer — and can I find 100 of them?"

Not "who might want this" but "who MUST have this." If you can't name 100 specific people or companies, your market understanding is too vague.

3. "What's my magic moment — and do users reach it?"

The single moment where a user first experiences the core value. For Superhuman: sending their first email and feeling the speed. For Airbnb: seeing the first professional photos of an apartment. Measure what percentage of new users reach this moment. Optimize ruthlessly for it.

4. “Would my best users riot if I took this away?”

Not “be sad” — riot. If the answer is no, you have a nice-to-have, not a must-have.

For Student Teams: The Lightweight Version

You probably don’t have enough users yet for a statistically significant survey. That’s OK. Use this adaptation:

After 10 customer conversations (Mom Test style), count: - How many described the problem unprompted? → Need 7+/10 - How many have an active workaround? → Need 5+/10 - How many asked when they can use it? → Need 3+/10 - How many offered to pay or pre-order? → Need 1+/10

After your first 20 users, run the Sean Ellis survey. 40 respondents gives you directionally correct results. Don’t wait for 1,000.

Common Mistakes

Surveying too broadly. Only survey people who’ve actually used the core product, at least twice. New signups who haven’t activated don’t count.

Ignoring the “Somewhat Disappointed” group. These aren’t failures — they’re your best clue. Half of them are one feature or one positioning change away from “Very Disappointed.”

Building for “Not Disappointed” users. This is the most seductive trap. They give specific, articulate feedback. It sounds smart. But they were never your customer, and building for them pulls you away from your actual value.

Measuring once and moving on. PMF isn’t binary. It’s a score that moves. Measure it every 2-4 weeks and track the trajectory.

Sources & Further Reading

- Superhuman’s PMF Engine — First Round Review
- Sean Ellis PMF Survey Tool

- Sequoia Arc PMF Framework
- Sequoia Arc Terrifying Questions